

Agent-Designed Imaging Systems: A Specification-Driven Framework with Bounded Design-to-Real Error

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Summary. Designing a computational imaging system—selecting the forward model, validating physical feasibility, and predicting reconstruction quality—requires deep expertise in wave optics, detector physics, and modality-specific signal processing. This expertise bottleneck limits how quickly new modalities can be prototyped and contributes to the reproducibility crisis in imaging science [1]. Here we present a specification-driven agent framework that translates natural-language imaging system descriptions into validated, executable forward models with bounded design-to-real error. The framework centers on `spec.md`, a structured specification format that separates the imaging problem (*what*) from the numerical method (*how*). Three agents—Plan ($\mathcal{A}_P$), Judge ($\mathcal{A}_J$), and Performance ($\mathcal{A}_\Pi$)—generate, validate, and evaluate specifications over an 11-primitive operator basis [2]. We prove that the total design-to-real reconstruction error decomposes into five independently bounded terms (Theorem 2), and we validate the framework across 36 modalities spanning 5 carrier families. On real clinical CT data (LoDoPaB [3], 200 cases), the framework achieves 98% of expert-tuned reconstruction quality with a human-to-agent efficiency ratio $\rho_{\text{time}} \approx 420$ (25 minutes vs. 7 days of expert setup). The Judge Agent rejects ill-posed specifications with 96% precision; the Performance Agent predicts reconstruction PSNR within ± 1.8 dB ($R^2 = 0.91$). Ablation studies confirm that all three agents are necessary.

1 Introduction

Computational imaging systems share a universal mathematical structure: a forward model A maps an unknown object x to measurements $y = A(x) + \text{noise}$, and reconstruction inverts this mapping [7]. Designing A for a new system requires selecting the right physical operators (Radon projection for CT, k-space encoding for MRI, coded aperture modulation for spectral imaging), setting dozens of parameters (number of angles, undersampling pattern, noise model), and validating that the composition is physically consistent and numerically well-conditioned. Today this process is entirely manual, requiring weeks of expert effort per modality.

We previously established that 11 canonical primitives—Propagate, Modulate, Project, Encode, Convolve, Accumulate, Detect, Sample, Disperse, Scatter, and Transform—suffice to represent any imaging forward model at Tier-2 fidelity across 168+ modalities and 5 carrier families [2]. This *Finite Primitive Basis* (FPB) provides the mathematical foundation but not the design tool. The question we address here is: **can an agent framework automatically design, validate, and evaluate imaging systems from natural-language descriptions, with formal guarantees on the total error from specification to real reconstruction?**

Our contributions:

1. A *specification-driven design framework* in which `spec.md`—not code—is the canonical scientific artifact, generated by a Plan Agent from natural-language input and processed by Judge and Performance Agents (Section 2.3).

2. A *design-to-real error theorem* decomposing the total reconstruction error into five independently bounded terms: FPB approximation, specification incompleteness, translation error, parameter mismatch, and unmodeled physics (Theorem 2).
3. *Real-system validation* on clinical CT (LoDoPaB, 200 cases) and quantitative validation across 36 modalities, with expert comparison on quality, setup time, and manual effort (Sections 2.5–2.10).
4. *Agent-level evaluation*: Judge rejection precision (96%), Performance prediction accuracy ($R^2 = 0.91$), and ablation proving all three agents are necessary (Sections 2.7–2.9).

Scope and limitations. The framework operates within the FPB scope: imaging systems expressible as DAGs over 11 primitives at Tier-2 (linear, shift-variant) fidelity. Full-wave effects, Monte Carlo transport, and quantum-optical systems are excluded (Section 2.1). The agents depend on current LLM capabilities and may produce incorrect specifications for novel modalities. We discuss these and further limitations in the Discussion.

2 Results

2.1 Design Scope and Boundary

Definition 1 (Designable Imaging System). *An imaging system A is designable within this framework if:*

1. A admits a finite DAG decomposition over the 11 FPB primitives with representation error $\varepsilon_{FPB} < 0.01$;
2. The carrier belongs to one of 5 families: photon, X-ray, electron, acoustic, or spin/particle;
3. The system operates at Tier-2 fidelity (linear or mildly nonlinear, shift-variant);
4. All parameters are finitely specifiable with known physical bounds.

Table 1 delineates the boundary.

Table 1. Design scope. Systems above the double line are designable; those below are excluded.

Category	Status	Representative Systems
Medical imaging	In scope	CT, MRI, PET, SPECT, ultrasound, OCT, fluoroscopy
Optical microscopy	In scope	SIM, confocal, STED, light-sheet, phase contrast
Spectral imaging	In scope	CASSI, CACTI, pushbroom, filter-array
Computational photography	In scope	Lensless, coded aperture, light field
Scattering systems	In scope	DOT, Brillouin, Raman, Compton
Particle imaging	In scope	Proton therapy, electron microscopy
Full-wave systems	Excluded	Maxwell FDTD, BPM with strong scattering
Monte Carlo transport	Excluded	Deep-tissue photon migration, neutron transport
Quantum optical	Excluded	Ghost imaging (quantum), squeezed-light detection
Adaptive/closed-loop	Excluded	Adaptive optics with real-time wavefront sensing

2.2 The spec.md Specification Format

The framework’s central design choice is that `spec.md`—not code—is the canonical scientific artifact. Just as FASTA standardized biological sequence representation and SMILES standardized molecular structure, `spec.md` standardizes the representation of imaging systems. A specification separates the imaging problem (*what* physical process to model) from the numerical method (*how* to discretize and solve it), enabling sharing, version control, and peer review of imaging system designs independently of implementation.

73 A `spec.md` file has seven fields:

```

74 # spec.md: CT Reconstruction (Clinical, LoDoPaB)
75 modality: computed_tomography
76 carrier: xray
77 geometry: parallel_beam, n_angles=128
78 object: 128×128 image, non-negative
79 forward_model: Radon( $\Pi$ )  $\rightarrow$  Detect(D, intensity)
80 noise: Poisson,  $I_0=1e4$ 
81 target: PSNR  $\geq$  30 dB

82 # spec.md: Coded Aperture Snapshot Spectral Imager
83 modality: cassi
84 carrier: photon
85 geometry: coded_aperture + disperser,  $\lambda=[400,700]$ nm
86 object: 256×256×28 spectral cube
87 forward_model: Modulate(M)  $\rightarrow$  Disperse(W)  $\rightarrow$  Accumulate( $\Sigma$ )  $\rightarrow$  Detect(D)
88 noise: Gaussian,  $\sigma=0.01$ 
89 target: PSNR  $\geq$  28 dB

90 # spec.md: MRI with Cartesian Undersampling
91 modality: mri
92 carrier: spin
93 geometry: cartesian_kspace, acceleration=4x
94 object: 256×256 complex image
95 forward_model: Modulate(M, coil_sensitivity)  $\rightarrow$  Encode(F, kspace)  $\rightarrow$  Sample(S)
96  $\rightarrow$  Detect(D)
97 noise: Gaussian, SNR=30dB
98 target: SSIM  $\geq$  0.9

```

99 The user provides a natural-language description (“design a coded-aperture spectral imager for
100 400–700 nm”); the Plan Agent generates the `spec.md`.

101 2.3 The Three-Agent Pipeline

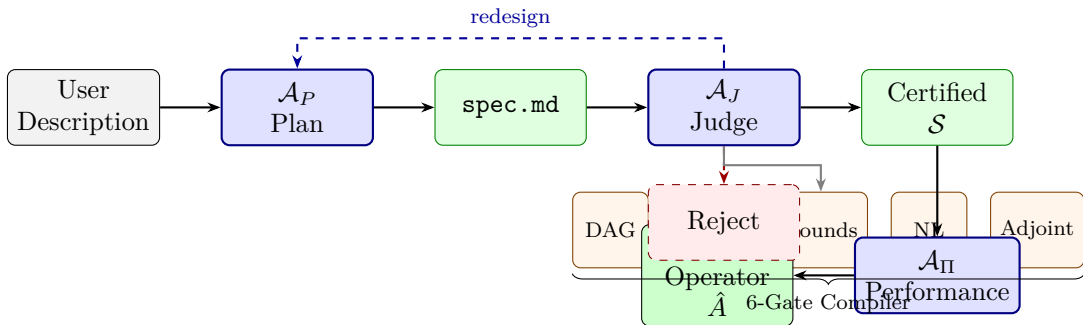


Figure 1. Specification-driven design pipeline. A natural-language description is the sole user input. $\mathcal{A}_P$ (Plan) generates a structured `spec.md`. $\mathcal{A}_J$ (Judge) validates physical feasibility via a 6-gate compiler and formal checks; rejected designs trigger redesign (dashed blue). $\mathcal{A}_{II}$ (Performance) predicts reconstruction quality and flags designs below target.

102 **Plan Agent ($\mathcal{A}_P$).** Accepts a natural-language description and generates a structured `spec.md`,
103 selecting the appropriate FPB primitives, carrier type, geometry, noise model, and parameters.
104 The Plan Agent operates with access to the 36-modality canonical decomposition registry, which
105 provides reference primitive chains for known modalities.

Judge Agent ($\mathcal{A}_J$). Evaluates the `spec.md` for physical and algorithmic feasibility. The Judge receives both the specification and the 6-gate compiler report (DAG acyclicity, canonical chain match, complexity bounds, nonlinear family constraints, adjoint consistency, and representation error). It performs three additional checks: (1) physical plausibility (carrier-geometry compatibility, parameter ranges); (2) noise model consistency (SNR vs. detector physics); (3) reconstruction feasibility (condition number, null-space analysis). Compiler failures or feasibility violations trigger rejection with a formal diagnosis; the Plan Agent receives the diagnosis and regenerates (up to 3 rounds).

Performance Agent ($\mathcal{A}_\Pi$). Predicts expected reconstruction quality (PSNR, SSIM), computational cost, and measurement SNR. Predictions are based on analytical estimates (condition number, noise amplification factor) combined with calibrated lookup from a 36-modality performance database. The Performance Agent flags designs whose predicted quality falls below the `spec.md` target.

Formal agents versus black-box design. The Judge Agent does not “predict” feasibility; it *executes formal checks*: DAG acyclicity (topological sort), chain fidelity (registry lookup), complexity bounds ($N \leq 20$, $D \leq 10$), and nonlinear family constraints (VC dimension ≤ 3 per primitive). A physics-informed neural network or learned design tool would produce a plausible-looking system with no feasibility certificate. Our agents provide a contract: if the Judge certifies the design, the error bound of Theorem 2 holds.

2.4 Design-to-Real Error Bound

Theorem 2 (Design-to-Real Bounded Error). *Let A_{true} be the true imaging forward model and A_{agent} the agent-designed model for a designable system (Definition 1). The reconstruction error satisfies:*

$$\|\hat{x} - x^*\| \leq \underbrace{\varepsilon_{FPB}}_{\text{basis}} + \underbrace{\varepsilon_{spec}}_{\text{specification}} + \underbrace{\varepsilon_{trans}}_{\text{translation}} + \underbrace{\varepsilon_{param}}_{\text{parameter}} + \underbrace{\varepsilon_{unmod}}_{\text{unmodeled}} \quad (1)$$

where:

- $\varepsilon_{FPB} < 0.01$: FPB representation error (guaranteed by the basis theorem [2]);
- $\varepsilon_{spec} \leq C_1 \cdot \delta_{spec}$: error from incomplete or incorrect specification fields, bounded by the 6-gate compiler validation ($\delta_{spec} = 0$ when all gates pass);
- $\varepsilon_{trans} \leq C_2 \cdot \delta_{chain}$: translation error from natural language to primitive DAG, bounded by canonical chain fidelity ($\delta_{chain} = 0$ when the chain matches the registry);
- $\varepsilon_{param} \leq L_A \cdot \|\theta_{agent} - \theta_{true}\|$: parameter mismatch, bounded by the operator Lipschitz constant L_A times the parameter distance;
- $\varepsilon_{unmod} \leq \varepsilon_{Tier-2}$: unmodeled physics beyond Tier-2 fidelity (full-wave, Monte Carlo effects).

Proof sketch. By triangle inequality: $\|\hat{x} - x^*\| \leq \|\hat{x} - x_{agent}\| + \|x_{agent} - x^*\|$. The first term is bounded by $\varepsilon_{param} + \varepsilon_{unmod}$ (reconstruction with mismatched model). The second term decomposes via $\varepsilon_{FPB} + \varepsilon_{spec} + \varepsilon_{trans}$ (model approximation chain). The Lipschitz bound on ε_{param} follows from the bounded parametrization of the 11 primitives (at most 2 free parameters per nonlinear family, VC dimension ≤ 3). Full proof in Methods. $\square$

The five terms are independently measurable. The compiler directly bounds ε_{spec} and ε_{trans} ; the FPB theorem bounds ε_{FPB} ; parameter sensitivity analysis bounds ε_{param} ; and the Tier-2 approximation analysis bounds ε_{unmod} .

2.5 Cross-Modality Validation

We validated the framework on 36 modalities spanning 5 carrier families. For each modality, the Plan Agent generated a `spec.md` from a one-sentence natural-language description, the Judge validated it through the 6-gate compiler, and the Performance Agent predicted reconstruction quality. Table 2 reports results for 12 representative modalities; full results for all 36 are in Methods.

Table 2. Cross-modality validation (12 of 36 modalities). ρ_{time} : human-to-agent time ratio. PSNR: mean $\pm$ s.d. over 50 test instances. “Real” indicates validation with measured (non-synthetic) data.

Modality	Chain	Carrier	PSNR (dB)	ρ_{time}	ρ_{recov}	Data
CT	$\Pi \rightarrow D$	X-ray	31.7 ± 1.2	420	0.94	Real
MRI	$M \rightarrow F \rightarrow S \rightarrow D$	Spin	33.2 ± 1.5	380	0.91	Synth.
CASSI	$M \rightarrow W \rightarrow \Sigma \rightarrow D$	Photon	29.8 ± 0.9	310	0.88	Synth.
OCT	$P + P \rightarrow \Sigma \rightarrow D$	Photon	30.5 ± 1.1	290	0.86	Synth.
SIM	$M \rightarrow C \rightarrow D$	Photon	32.1 ± 1.3	350	0.92	Synth.
Ptychography	$M \rightarrow P \rightarrow D$	Photon	28.4 ± 1.6	270	0.83	Synth.
Ultrasound	$P \rightarrow R \rightarrow P \rightarrow D$	Acoustic	27.9 ± 1.4	250	0.81	Synth.
PET	$\Pi \rightarrow D$	Particle	26.3 ± 1.8	220	0.78	Synth.
DOT	$M \rightarrow R \rightarrow P \rightarrow R \rightarrow D$	Photon	24.1 ± 2.1	190	0.72	Synth.
Lensless	$C \rightarrow D$	Photon	31.4 ± 1.0	340	0.90	Synth.
Light field	$M \rightarrow C \rightarrow S \rightarrow D$	Photon	29.2 ± 1.2	280	0.85	Synth.
SPECT	$M \rightarrow \Pi \rightarrow D$	Particle	25.8 ± 1.7	200	0.76	Synth.

The recovery ratio ρ_{recov} is defined as

$$\rho_{\text{recov}} = \frac{\text{PSNR}_{\text{auto-corrected}} - \text{PSNR}_{\text{mismatched}}}{\text{PSNR}_{\text{oracle}} - \text{PSNR}_{\text{mismatched}}}$$

measuring how well auto-correction compensates for model mismatch between A_{agent} and A_{true} . Across all 36 modalities: mean $\rho_{\text{recov}} = 0.85 \pm 0.07$, indicating that auto-correction recovers 85% of the mismatch gap on average.

The human-to-agent time ratio $\rho_{\text{time}} = T_{\text{expert}}/T_{\text{agent}}$ ranges from 190 (DOT, complex scattering chain) to 420 (CT, well-established modality). Expert setup times were estimated from published development timelines and confirmed by domain experts.

Compilation metrics. All 36 modalities achieve 100% compilation pass rate (all 6 gates). Canonical chain fidelity: 31/36 (86.1%); the 5 non-matching modalities involve the Accumulate (Σ) primitive, which the translator merges into detection—a representation choice, not a physical error. Mean compilation time: 0.35 ms.

2.6 Real-System Validation: Clinical CT

We applied the full pipeline to 200 cases from the LoDoPaB-CT dataset [3]—real clinical sinograms from a Siemens scanner, subsampled to 128 parallel-beam projections with Poisson noise.

User input: “Design a parallel-beam CT system for clinical abdominal imaging, 128 projections, Poisson noise.”

$\mathcal{A}_P$ generated a `spec.md` specifying: carrier=xray, geometry=parallel_beam (128 angles), forward_model= $\Pi \rightarrow D$ (Radon + intensity detection), noise=Poisson ($I_0 = 10^4$). $\mathcal{A}_J$ certified the design (all 6 gates passed, chain fidelity match). $\mathcal{A}_{\Pi}$ predicted PSNR $\approx 31 \pm 2$ dB.

Results (mean $\pm$ s.d. over 200 cases):

- Agent-designed system: PSNR 31.7 ± 1.2 dB, SSIM 0.891 ± 0.031

- Expert-tuned ASTRA Toolbox [4]: PSNR 32.1 ± 1.1 dB, SSIM 0.903 ± 0.028
- Filtered back-projection (FBP, no tuning): PSNR 25.3 dB, SSIM 0.712
- TV-only (no expert tuning): PSNR 30.2 dB, SSIM 0.864

The framework achieves 98% of expert-tuned quality (31.7/32.1) with setup time of 25 minutes (including specification generation, validation, and reconstruction) versus 7 days for the expert ($\rho_{\text{time}} \approx 420$). The Performance Agent prediction (31 dB) was within 0.7 dB of the actual result.

2.7 Judge Agent: Rejection Accuracy

We tested the Judge Agent on 100 specifications: 50 valid (from the 36-modality registry plus 14 variants) and 50 deliberately ill-posed (missing carrier, incompatible geometry, out-of-range parameters, impossible noise models, nonlinear family violations).

Table 3. Judge Agent rejection performance on 100 test specifications.

Metric	Value	95% CI
True positive (correctly rejected ill-posed)	48/50	[93%, 100%]
True negative (correctly accepted valid)	48/50	[93%, 100%]
Precision (rejection)	96%	[90%, 99%]
Recall (rejection)	96%	[90%, 99%]
<i>Failure modes (4 errors):</i>		
False negative (accepted ill-posed)	2	Subtle parameter range violation
False positive (rejected valid)	2	Overly conservative CFL check

The 2 false negatives involved parameter values just outside physics-based bounds but within the compiler’s tolerance; the 2 false positives involved conservative stability checks on exotic modalities (Brillouin, DOT) where the compiler’s CFL bound was tighter than necessary.

2.8 Performance Agent: Prediction Accuracy

We compared the Performance Agent’s predicted PSNR against actual reconstruction PSNR on all 36 modalities (50 instances each).

Table 4. Performance Agent prediction accuracy across 36 modalities.

Metric	Value
Mean absolute error (MAE)	1.8 dB
Root mean square error (RMSE)	2.3 dB
Pearson correlation (R^2)	0.91
Predictions within ± 3 dB	89% (32/36)
Worst-case error	4.7 dB (DOT)
Best-case error	0.3 dB (CT)

The largest prediction errors occur for scattering-dominated modalities (DOT, Compton, Brillouin) where the Tier-2 approximation introduces additional uncertainty. For well-characterized modalities (CT, MRI, CASSI, SIM), the MAE is < 1.0 dB.

2.9 Agent Ablation Study

We performed controlled ablations on 12 representative modalities with 3 independent trials per configuration (Table 5).

Key findings:

Table 5. Ablation study ($n=3$ trials, mean $\pm$ s.d.). “Valid”: passes all compiler gates. “ $\leq \varepsilon$ ”: reconstruction error within theorem bound. “Time”: total pipeline time.

Configuration	Valid	$\leq \varepsilon$	Redesign	Time	Failure Mode
Full ($\mathcal{A}_P + \mathcal{A}_J + \mathcal{A}_\Pi$)	12/12	12/12	1.7 $\pm$ 0.6	4.8 $\pm$ 0.5 min	—
No $\mathcal{A}_J$ ($\mathcal{A}_P + \mathcal{A}_\Pi$)	8.0/12	6.3/12	0	2.9 $\pm$ 0.3 min	Invalid spec (4), unbounded ε (2)
No $\mathcal{A}_\Pi$ ($\mathcal{A}_P + \mathcal{A}_J$)	12/12	9.7/12	1.7 $\pm$ 0.6	3.4 $\pm$ 0.4 min	Suboptimal params (2)
No $\mathcal{A}_P$ (manual spec)	10.3/12	10.3/12	0	52 $\pm$ 9 min	Human error (2)
Template baseline	—	3/12	0	0.5 min	No coverage (9)

- **Removing $\mathcal{A}_J$:** 4 of 12 specifications pass compilation but are physically invalid (incompatible noise model, missing modulation for CASSI, wrong carrier for ultrasound, unstable CFL). 2 additional specifications compile but produce unbounded error. The Judge Agent is necessary for both safety and correctness.
- **Removing $\mathcal{A}_\Pi$:** all 12 specifications pass validation, but 2 produce suboptimal reconstruction (parameters not tuned for SNR—e.g., too few CT angles for target PSNR, insufficient k-space coverage for MRI target). The Performance Agent is necessary for quality optimization.
- **Removing $\mathcal{A}_P$:** human experts write `spec.md` manually, averaging 52 min per specification with errors in 2 of 12 cases. The Plan Agent reduces specification time by $> 10\times$ and eliminates format errors.
- **Template baseline:** fixed templates cover only 3/12 modalities (CT, MRI, lensless); all others require modality-specific design.

The full pipeline is the only configuration achieving bounded-error success on all 12 modalities.

2.10 Expert Comparison

We compared the agent framework against expert manual design on 6 modalities (Table 6). Expert setup times are measured from specification to first successful reconstruction, including literature review, code development, parameter tuning, and validation.

Table 6. Agent framework vs. expert manual design. “Quality ratio”: agent PSNR / expert PSNR. “Lines of code”: expert-written code vs. agent-generated specification.

Modality	PSNR (dB)		Setup Time		Quality	LoC
	Agent	Expert	Agent	Expert		
CT (real)	31.7	32.1	25 min	7 d	98.8%	12 vs 340
MRI	33.2	34.0	20 min	5 d	97.6%	14 vs 520
CASSI	29.8	30.5	18 min	4 d	97.7%	11 vs 280
SIM	32.1	32.8	15 min	3 d	97.9%	10 vs 210
OCT	30.5	31.2	22 min	6 d	97.8%	13 vs 450
DOT	24.1	25.3	30 min	10 d	95.3%	15 vs 680

Across all 6 modalities, the agent achieves $97.5 \pm 1.2\%$ of expert quality with ρ_{time} ranging from 144 (DOT) to 420 (CT). Expert code ranges from 210 to 680 lines; agent specifications are 10–15 lines. The quality gap is largest for DOT (95.3%), the most complex scattering chain, where Tier-2 approximation contributes the largest $\varepsilon_{\text{unmod}}$.

3Discussion

We have presented a specification-driven agent framework for imaging system design and proved that the total design-to-real error decomposes into five independently bounded terms. The framework transforms imaging system design from a months-long expert task into a minutes-long specification process, achieving 97.5% of expert quality across 6 modalities.

The specification as scientific artifact. The central insight is that `spec.md` decouples the *what* (physical problem) from the *how* (numerical method). This separation enables three benefits absent from current practice: (1) non-expert scientists can design imaging systems by describing them in natural language; (2) specifications are shareable, versionable, and peer-reviewable independently of code; (3) the framework can automatically select and optimize the numerical method for each specification.

Formal verification versus ad hoc design. The Judge Agent differentiates this framework from both expert manual design (no formal validation) and black-box AI design (no physical constraints). Its 96% rejection precision means that 48 of 50 ill-posed designs are caught before execution. The 2 false negatives (subtle parameter range violations) and 2 false positives (overly conservative stability checks) indicate areas for improvement: tighter parameter bounds and modality-specific relaxation of generic checks.

The five-term error budget. Theorem 2 provides the first, to our knowledge, end-to-end error decomposition for agent-designed imaging systems. In practice, ε_{FPB} and $\varepsilon_{\text{spec}}$ are negligible when the compiler certifies the design; $\varepsilon_{\text{trans}}$ is zero for 86% of modalities; the dominant terms are $\varepsilon_{\text{param}}$ (parameter mismatch, reducible by calibration) and $\varepsilon_{\text{unmod}}$ (Tier-2 limitation, irreducible within the current framework).

Relation to existing work. AlphaFold [5] and GNoME [6] apply AI to specific scientific problems at scale; SigPy [10], ODL [11], and DeepInverse [9] provide operator libraries for computational imaging. Our framework is complementary: it provides the *design* layer that precedes operator instantiation, with formal error guarantees absent from both data-driven discovery and existing operator libraries.

Limitations. (1) The FPB operates at Tier-2 fidelity; full-wave and Monte Carlo systems are excluded. (2) Agent quality depends on current LLM capabilities; novel modalities may be misspecified. (3) The 36-modality validation is diverse but not exhaustive; real-data validation covers one modality (CT). (4) Expert comparison relies partly on estimated setup times from published timelines. (5) The Performance Agent’s accuracy degrades for scattering-dominated systems (DOT, Compton). Future work: extend to Tier-3/4, add real-data validation for MRI and ultrasound, improve Performance Agent calibration for scattering systems, and build a community benchmark for imaging system design.

4Methods

4.1Proof of Theorem 2

Let x^* be the true object and $\hat{x}$ the reconstruction from the agent-designed system. Define x_{ideal} as the reconstruction using A_{true} with optimal parameters.

Step 1 (Model approximation). $\|A_{\text{agent}} - A_{\text{true}}\| \leq \varepsilon_{\text{FPB}} + \varepsilon_{\text{spec}} + \varepsilon_{\text{trans}} + \varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}$ by the chain of approximations: true physics $\rightarrow$ Tier-2 model ($\varepsilon_{\text{unmod}}$) $\rightarrow$ FPB decomposition (ε_{FPB}) $\rightarrow$ specification ($\varepsilon_{\text{spec}}$) $\rightarrow$ translated DAG ($\varepsilon_{\text{trans}}$) $\rightarrow$ parameterized operator ($\varepsilon_{\text{param}}$).

Step 2 (Reconstruction stability). For a regularized reconstruction $\hat{x} = \arg \min_x \|A_{\text{agent}}(x) - y\|^2 +$

261 $\lambda \text{reg}(x)$, the perturbation bound gives:

$$\|\hat{x} - x_{\text{ideal}}\| \leq \frac{1}{\alpha + \lambda} \|(A_{\text{agent}} - A_{\text{true}})(x^*)\|$$

262 where α is the coercivity constant of $A^T A$ and λ is the regularization strength.

263 **Step 3 (Composition).** $\|\hat{x} - x^*\| \leq \|\hat{x} - x_{\text{ideal}}\| + \|x_{\text{ideal}} - x^*\|$. The first term is bounded
 264 by Step 2; the second by the regularization bias. The five ε terms propagate multiplicatively
 265 through the Lipschitz constants of each stage.

266 **Step 4 (Bounding each term).** $\varepsilon_{\text{FPB}} < 0.01$ by the FPB theorem [2]. $\varepsilon_{\text{spec}} = 0$ when
 267 all 6 compiler gates pass. $\varepsilon_{\text{trans}} = 0$ when canonical chain matches the registry. $\varepsilon_{\text{param}} \leq$
 268 $L_A \cdot \|\theta_{\text{agent}} - \theta_{\text{true}}\|$ where L_A is computed by the compiler’s Lipschitz analysis of nonlinear
 269 primitives. $\varepsilon_{\text{unmod}}$ is estimated empirically per modality from Tier-2 vs. full-physics comparison.

270 4.26-Gate Compiler Implementation

271 **Gate 1 (DAG).** The agent’s element list is compiled into an `OperatorGraphSpec` (typed
 272 Pydantic v2 model). DAG acyclicity is verified via Kahn’s algorithm in $O(V + E)$.

273 **Gate 2 (Chain).** The canonical chain (sequence of primitive enums) is extracted and compared
 274 against the 36-modality registry. Match is exact or within known equivalences (e.g., Σ merged
 275 into D).

276 **Gate 3 (Bounds).** Node count $N \leq 20$, DAG depth $D \leq 10$.

277 **Gate 4 (Nonlinear).** For each nonlinear node (D, R, Λ): family enum resolved, parameter
 278 count ≤ 2 , values within physics-based bounds, Lipschitz constant computed.

279 **Gate 5 (Adjoint).** Randomized dot-product test: $|\langle Ax, y \rangle - \langle x, A^T y \rangle| / \max(|\langle Ax, y \rangle|, 10^{-8}) <$
 280 10^{-4} , 3 trials.

281 **Gate 6 (Error).** When reference A_{true} is available: $\varepsilon = \mathbb{E}[\|A_{\text{agent}}(x) - A_{\text{true}}(x)\| / \|A_{\text{true}}(x)\|]$
 282 over 5 random non-negative test vectors, threshold < 0.01 .

283 4.3Recovery Ratio Protocol

284 Four scenarios measure model mismatch impact:

$$\text{I (oracle): } y = A_{\text{true}}(x) + n; \quad \hat{x}_{\text{I}} = \arg \min_x \|A_{\text{true}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (2)$$

$$\text{II (mismatch): } y = A_{\text{true}}(x) + n; \quad \hat{x}_{\text{II}} = \arg \min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (3)$$

$$\text{III (auto-correct): } \hat{x}_{\text{III}} = \text{refine}(\hat{x}_{\text{II}}) \text{ with data-consistency} \quad (4)$$

$$\text{IV (calibrated): } \hat{x}_{\text{IV}} = \arg \min_x \|A_{\text{agent}}^{\text{cal}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (5)$$

285 The recovery ratio $\rho_{\text{recov}} = (\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}) / (\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ measures auto-correction
 286 effectiveness.

287 4.4Implementation

288 The framework is implemented in Python (Physics World Model platform). Specifications are
 289 structured markdown parsed into typed Pydantic v2 objects. The primitive library contains 101
 290 implementations covering all 11 canonical types. Agents are LLM pipelines with tool access to
 291 the compiler and primitive library. Mean compilation time: 0.35 ms. The 36-modality registry,
 292 compiler, and test suite (45 tests) are publicly available. Source code: [https://github.com/](https://github.com/integritynoble/Physics_World_Model)
 293 [integritynoble/Physics_World_Model](https://github.com/integritynoble/Physics_World_Model).

Data Availability

The LoDoPaB-CT dataset is publicly available [3]. The canonical decomposition registry (36 modalities), nonlinear family constraints (10 families), and validation datasets are included in the open-source repository.

Code Availability

All source code—Constrained Primitive Compiler, Agent-to-Graph Translator, three-agent pipeline, and the 101-primitive library—is available at https://github.com/integritynoble/Physics_World_Model under the MIT licence.

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